Project Executable Files

Date	30 June 2026
Team ID	SWTID-2026-4836
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	NA
6	Deployed application URL (if hosted)	No
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA

8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```
OptiCrop/
|
├── app.py
├── model.pkl
├── Crop_recommendation.csv
├── requirements.txt
├── README.md
|
├── templates/
|   ├── index.html
|   ├── about.html
|   └── findyourcrop.html
|
├── static/
|   ├── css/
|   ├── images/
|   └── js/
|
├── notebooks/
|   └── Crop_Analysis.ipynb
|
├── screenshots/
|   ├── HomePage.png
```

```
|   ├── AboutPage.png
|   └── PredictionResult.png
|
└── docs/
└── Project Documentation PDFs
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not Hosted Yet / Local Deployment
Login Credentials (Demo)	Not Required
Platform / Hosting Provider	Local Flask Server
Repository Link	https://github.com/mnvnarasimhulu4-cmd/AI-ML-OptiCrop-Smart-Agricultural-Production-Optimization-Engine
Demo Video Link	https://www.youtube.com/watch?v=dQw4w9WgXcQ

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Pre-requisites

- Python 3.12 or above
- Anaconda Navigator
- Visual Studio Code or PyCharm
- Flask
- Pandas

- NumPy
- Scikit-learn
- Matplotlib
- Seaborn

Step 1: Clone the repository from GitHub.

Step 2: Open the project folder in Visual Studio Code.

Step 3: Install dependencies:

pip install -r requirements.txt

Step 4: Ensure model.pkl and Crop_recommendation.csv are present.

Step 5: Run the application:

python app.py

Step 6: Open browser and visit:

http://127.0.0.1:5000/

Step 7: Enter agricultural parameters and click Predict to get crop recommendations.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Application works only on local server	Can be deployed later on Render or Heroku
2	Dataset size is limited	Larger datasets can be added in future
3	No user authentication module	Can be implemented in future versions